MERNA MAMDOOH

PORTFOLIO / PROJECT 01

Super Fit Gym

Gym performance analytics and management system

Merna Mamdooh

Junior Data Analyst candidate / Cairo, Egypt

I turn operational data into clear measures of branch performance, cash collection and customer acquisition. My background in business development helps me connect analysis to practical business questions.

6	1,200	38,774
demo branches	unique members	attendance records

The business question

How can a gym network compare branches, monitor membership activity and distinguish sales value from actual cash collected?

My project scope

Requirements, data structure, KPI definitions, dashboard design, branch exploration, upload workflow and role-based access. This case study presents the analytical reasoning and independently calculated results from the project workbook.

Independent portfolio project using fully synthetic data and fictional branch locations.
Reporting period: 1 October 2025 - 30 September 2026. Currency: EGP.

DATA FOUNDATIONS

Model the business correctly

Separate people, contracts, cash transactions and visits before calculating KPIs.

Dataset	Rows	What one row represents
Branches / Plans	6 / 6	One branch / one membership plan
Employees / Members	72 / 1,200	One employee / one unique member
Leads	3,400	One prospect
Memberships	1,641	One membership contract
Payment schedule	2,741	One scheduled installment
Payments	2,573	One collection or refund transaction
Attendance	38,774	One gym visit
Sales targets	144	One salesperson-month target

Join strategy

Member_ID connects members to contracts and visits. Membership_ID connects each contract to payments and installments. Branch_ID supports sales comparisons, while Visit_Branch_ID identifies where attendance actually occurred.

Avoid double counting

Aggregate payments by Membership_ID before joining to contracts. Joining raw payments and raw attendance directly would multiply rows. Count members with distinct Member_ID, rather than counting membership contracts.

METRIC DESIGN AND VALIDATION

Separate sales from cash

Snapshot: 30 September 2026. Calculations below use the full reporting dataset.

EGP 14.318m
adjusted contract value

EGP 13.555m
net cash collected

Measure	Definition	Result
Adjusted value	Contract value less credits	EGP 14,317,625
Net cash	Collections + signed refunds	EGP 13,554,740
Outstanding	Adjusted value - net cash	EGP 762,885
Overdue	Unpaid amount already due	EGP 247,216
Cash / adjusted value	Net cash / adjusted value	94.7%
Lead conversion	Converted leads / all leads	35.3%

Checks performed for this case study

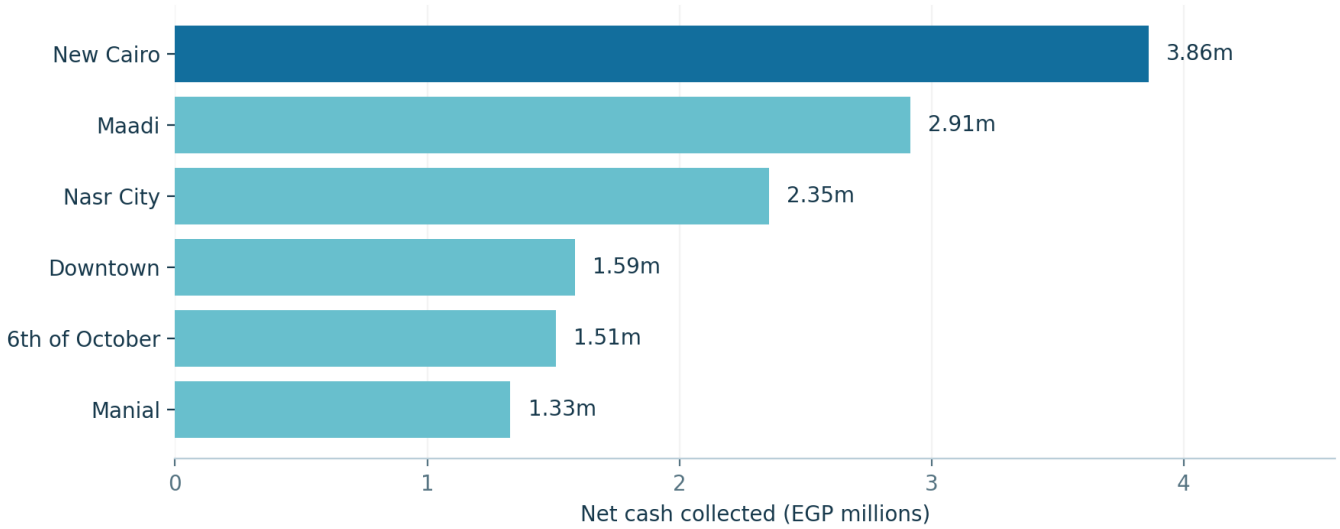
Nine entity tables: no blank or duplicated primary keys. Seven core relationships: no unmatched foreign keys. Payment totals reconcile to membership-level net cash. Adjusted value reconciles to net cash plus outstanding, and overdue does not exceed outstanding.

Collections: EGP 13,910,540. Refunds: EGP -355,800.
Adjusted contract value is a sales measure, not recognized accounting revenue. The 94.7% measure includes future installments in its base.

BRANCH PERFORMANCE

Where cash is collected

Full-period comparison. Branches are fictional training locations.



50.0%

of net cash comes from the top two branches

EGP 65,460

Maadi overdue balance

Finding

New Cairo leads net cash collection at EGP 3,859,435. Together, New Cairo and Maadi account for 50.0% of the network total.

Decision to investigate

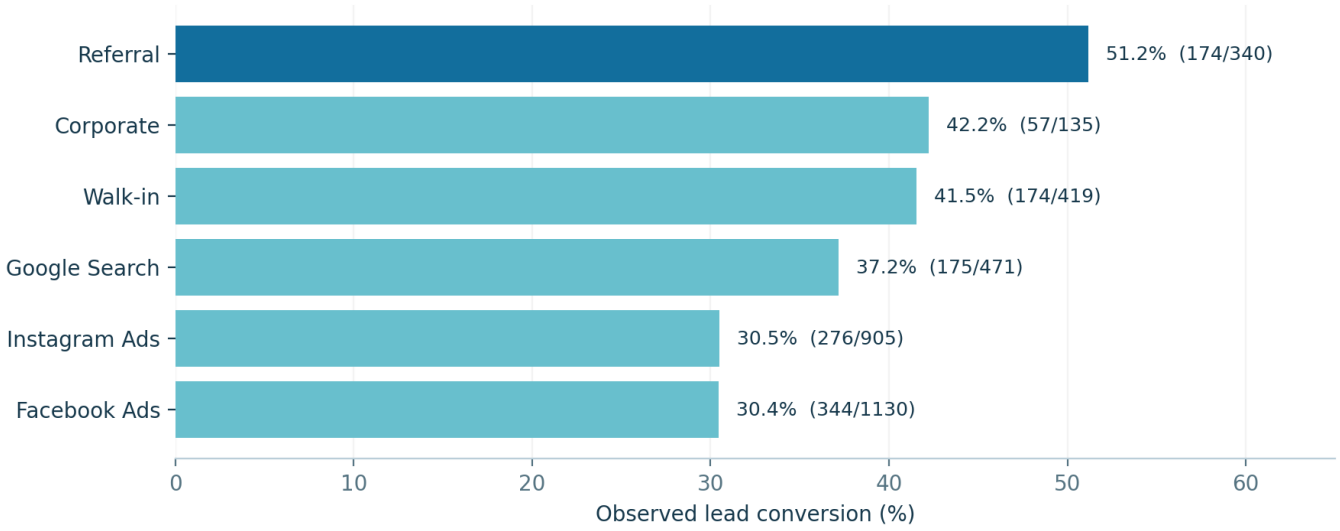
Maadi has EGP 65,460 overdue, close to New Cairo at EGP 65,963 despite a smaller sales base. Review overdue balances relative to adjusted contract value: 2.10% in Maadi versus 1.62% in New Cairo.

Recommended action: prioritize overdue installment follow-up, then compare aging and payment schedules. Branch size and contract mix may explain differences; these totals do not establish a cause.

ACQUISITION ANALYSIS

Volume and conversion differ

Observed outcomes for all 3,400 leads created in the reporting period.



51.2%

Referral conversion: 174 of 340 leads

344

Facebook Ads conversions from 1,130 leads

Finding

Referrals have the highest observed conversion rate. Facebook Ads generate the most conversions in absolute terms, with a lower conversion rate of 30.4%. A channel ranking changes depending on whether the goal is efficiency or volume.

Recommendation and analytical limits

Test a referral initiative and monitor both converted volume and conversion rate. Review paid-channel lead quality and follow-up. Marketing spend is unavailable, so this analysis cannot rank ROI or cost per acquisition.

Open leads remain in the denominator. This is a descriptive snapshot, not a mature-cohort conversion study or evidence that one channel causes better results.

FROM ANALYSIS TO USE

A repeatable reporting workflow

Translate the data model into an interface that supports daily operational review.

System component	Business purpose
Excel update workflow	Refresh operational data and dependent dashboard views
Branch and area filters	Explore local performance and map locations
Members and memberships	Review people separately from contract history
Sales and collections	Compare contract value, cash and unpaid balances
Attendance views	Explore activity and usage by visit branch
User roles	Separate CEO, Data Manager and Viewer responsibilities

Implementation context

The project guide documents a TypeScript dashboard, API routes, D1 for accounts and sessions, and R2 for dataset storage. This portfolio adds reproducible Python calculations and charts from the Excel workbook. Figures shown here are analytical exhibits, not application screenshots.

What the analysis supports

Prioritize overdue balances, compare branch cash performance, and assess acquisition channels using both conversion volume and rate. Evaluate recommendations with future measurements before claiming operational improvement.

Source: Super_Fit_Gym_Synthetic_Dataset.xlsx, version 2; Super Fit Gym build guide, 28 October 2026. All project data is synthetic. No measured commercial impact is claimed.